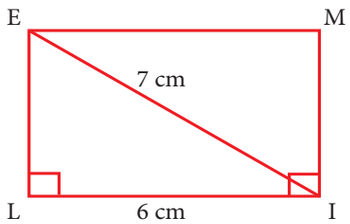
5.15.2 Construction of a rectangle when a side and a diagonal are given

Example 5.39

Construct a rectangle LIME with LI = 6 cm and IE = 7 cm. Also find its area.

Solution:

Given: LI = 6 cm and IE = 7 cm



Rough diagram

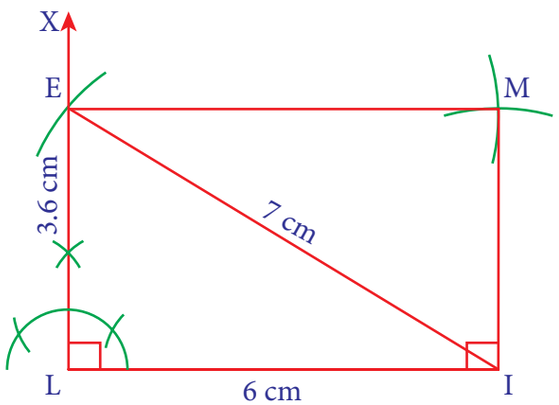


Fig. 5.58

Steps:

1. Draw a line segment LI = 6 cm.
2. At L, construct $LX \perp LI$.
3. With I as centre, draw an arc of radius 7 cm and let it cut LX at E.
4. With I as centre and LE as radius draw an arc. Also, with E as centre and LI as radius draw an another arc. Let them cut at M.
5. Join IM and EM.
6. LIME is the required rectangle.

Calculation of Area:

Area of rectangle LIME $= l \times b \; sq.units$

$= 6 \times 3.6 = 21.6 \; sq.cm$

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